Capacitor Charging and a Power On Reset for an Active Low Microcontroller

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***Abstract*—This report looks at capacitor charging in an RC circuit and shows how the same circuit can be used as a simple power-on-reset circuit for a microcontroller. The charging curve is compared for different resistor values. The reset release time is also calculated for an active-low reset pin.**

I. Introduction

An RC circuit is made from a resistor and a capacitor. When power is applied, the capacitor does not charge instantly. Its voltage rises over time.

If a capacitor C charges through a resistor R from a constant supply voltage V_0, the capacitor voltage is

$$V(t) = V_0 \left(1 - e^{-t/RC}\right). \tag{1}$$

The product RC is called the time constant,

$$\tau = RC. \tag{2}$$

The time constant gives an idea of how fast the capacitor charges. A larger resistance or capacitance makes the charging process slower.

At one time constant,

$$V(\tau) = V_0 \left(1 - e^{-1}\right) \approx 0.632 V_0. \tag{3}$$

For a supply voltage of 5.0 V,

$$V(\tau) \approx 0.632(5.0) \approx 3.16 \text{ V}. \tag{4}$$

Figure 1 compares the charging curve for four different resistor values while the capacitor stays at 22 pF.

The $10\,\mathrm{k\Omega}$ resistor charges the capacitor the fastest. The larger resistor values take more time to reach the same voltage. This agrees with the time constant equation because increasing R increases τ.

II. Power On Reset

The RC circuit can also be used as a simple power-on-reset circuit. This can help keep a microcontroller in reset for a short time when the device first turns on.

The circuit used here is connected to an active-low reset pin. An active-low reset means the microcontroller stays in reset while the reset pin voltage is low.

At power-up, the capacitor starts discharged. Because of this, the reset node is initially close to ground. As the capacitor charges through the resistor, the voltage at the reset pin increases.

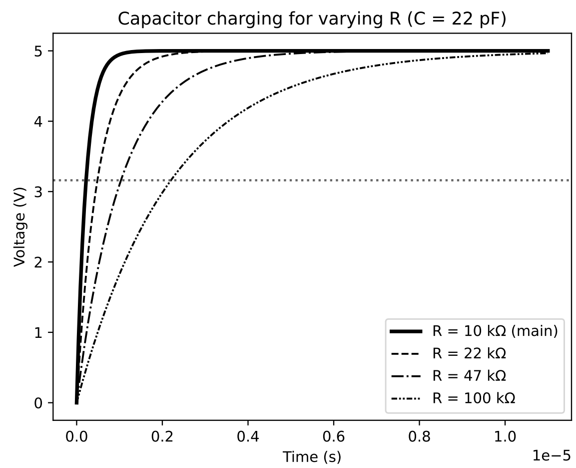

Fig. 1. Capacitor charging for different resistor values with $C = 22\,\mathrm{pF}$.

The reset is released when the voltage reaches the microcontroller's input-high threshold V_{IH}.

Starting from the charging equation and setting the capacitor voltage equal to V_{IH},

$$V_{IH} = V_0 \left(1 - e^{-t_{\text{release}}/RC}\right). \tag{5}$$

Dividing by V_0 gives

$$\frac{V_{IH}}{V_0} = 1 - e^{-t_{\text{release}}/RC}. \tag{6}$$

Rearranging,

$$e^{-t_{\text{release}}/RC} = 1 - \frac{V_{IH}}{V_0}. \tag{7}$$

Taking the natural logarithm,

$$-\frac{t_{\text{release}}}{RC} = \ln\left(1 - \frac{V_{IH}}{V_0}\right). \tag{8}$$

The release time is therefore

$$t_{\text{release}} = -RC \ln\left(1 - \frac{V_{IH}}{V_0}\right). \tag{9}$$

If the desired release time is already known, the equation can be rearranged to solve for R:

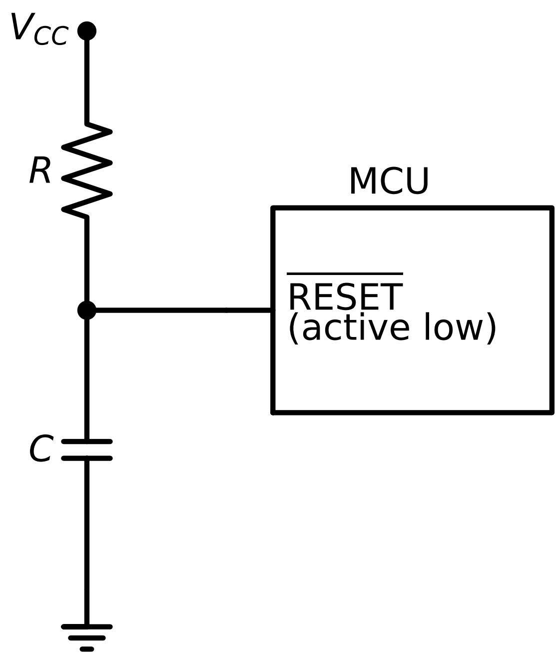

Fig. 2. RC power-on-reset circuit connected to an active-low reset pin.

$$R = \frac{-t_{\text{release}}}{C \ln\left(1 - \frac{V_{IH}}{V_0}\right)}. \tag{10}$$

III. Worked Example

For this example, the microcontroller threshold is

$$V_{IH} = 0.7V_0. \tag{11}$$

The supply voltage is

$$V_0 = 5.0 \text{ V}. \tag{12}$$

The required reset delay is

$$t_{\text{release}} = 10 \text{ ms}, \tag{13}$$

and the capacitor is

$$C = 100 \text{ nF}. \tag{14}$$

First,

$$\ln(1 - 0.7) = \ln(0.3) \approx -1.204. \tag{15}$$

Substituting the values into the resistor equation,

$$R = \frac{-(10 \times 10^{-3})}{(100 \times 10^{-9})(-1.204)} \approx 83\,\text{k}\Omega. \tag{16}$$

An $83\,\text{k}\Omega$ resistor with a 100 nF capacitor gives a reset delay of about 10 ms.

The 22 pF capacitor used in the charging plot is useful for showing the charging curve clearly, but it is too small for this reset timing example. A larger capacitor such as 100 nF gives a practical resistor value in the kilo-ohm range. This is why the capacitor value is changed for the power-on-reset design.

IV. Conclusion

The RC charging equation shows how the resistor and capacitor control the charging speed. Increasing the resistance increases the time constant and makes the capacitor charge more slowly.

The same charging behaviour can be used for a simple power-on-reset circuit. For the example used in this report, an $83\,\text{k}\Omega$ resistor and a 100 nF capacitor give a reset delay of about 10 ms.